

Walter Wang

Senior AI Engineer — Computer Vision & Agentic AI

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PROFILE

AI Engineer with 4+ years of commercial experience spanning Computer Vision, Agentic AI, and cloud-native backend engineering. At Imagr I designed, trained, and edge-deployed object-detection, segmentation, and classification models for real-time smart shopping-cart systems. More recently at ZEIL I architect and ship multi-agent, LLM-driven pipelines (Pydantic-AI, Gemini) on GCP that automate end-to-end workflows at scale, and built real-time full-stack products with React/Next.js and AWS. I take ideas from prototype to production and focus on outcomes and measurable value.

EXPERIENCE

AI Engineer

Jul 2025 – Present

ZEIL | Auckland, NZ

- **Auto-apply pipeline:** Designed an event-driven, multi-tenant pipeline that scrapes ATS job-application forms with headless browser automation, maps fields using AI structured outputs, and auto-fills applications at scale across multiple ATS vendors.
- **Company Intelligence Agent:** Built a provider-agnostic AI agent with custom web-search and scraping tools that, given a company name from a candidate's CV, autonomously retrieves live data — business activity, funding, tech stack, and news — and synthesises a structured summary to give recruiters instant context on a candidate's previous employer.
- **CV Processing & Candidate Ranking:** Developed an LLM pipeline that parses uploaded CVs, extracts structured fields, calculates weighted match scores against job requirements, and ranks candidates — reducing manual screening effort and improving shortlist quality.
- **Talent-seeker platform:** Full-stack LinkedIn outreach app with real-time messaging, relation management, and live event sync; cloud-native backend with managed database and real-time subscriptions.

AI Engineer (Contractor)

May 2025 – Jun 2025

Comply Copilot | Remote

- Built LLM pipelines to link council activities to compliance standards, filter recommended activities (~1,100 → ~200 via relevance scoring), and evaluate project plans against regulatory rules — with structured outputs, async concurrency, and fallback/retry for reliability.
- Cut token costs ~30% through deduplication, batching, and selective model choices; improved output quality via iterative prompt design and schema-enforced responses.

Artificial Intelligence Engineer

May 2022 – Apr 2025

Imagr | Auckland, NZ

- Designed, trained, and deployed Computer Vision models for smart shopping carts: **Object Detection** (YOLO, SSD, Faster R-CNN), **Segmentation** (U-Net, Mask R-CNN, SAM), **Classification** (MobileNet, EfficientNet, Inception); customised architectures and loss functions to suit real-world retail constraints — varied lighting, partial occlusion, and high SKU count.
- Full model lifecycle — collect data, train, deploy, monitor production metrics, identify failure modes, then close the loop: targeted data collection in scenarios where the model performed worst, re-annotation, and iterative fine-tuning to continuously raise accuracy.
- Self-hosted CVAT for annotation; trained a large model on an initial labelled dataset and used it to auto-annotate new data at scale — human reviewers corrected predictions and refined labels were fed back to fine-tune the model, producing a high-quality dataset that drove sustained accuracy gains while cutting annotation cost ~90% and time ~50%.
- Implemented native Bayer RAW image processing in the inference pipeline — decoding RAW sensor frames directly rather than converting to RGB first — eliminating a preprocessing step and improving end-to-end inference speed.
- Optimised models for **Google EdgeTPU** via quantisation: ~75% model size reduction, 2–4× inference speedup with minimal accuracy loss; deployed on Coral Dev Board with C++.

- Trained on Google Cloud TPUs with TFRecord datasets; implemented SORT for real-time multi-object tracking.

Intern — Salesforce Developer

Nov 2021 – Feb 2022

Davanti Consulting | Auckland, NZ

- Developed a CRM for a non-profit on Salesforce; gathered requirements, implemented features, tested, and delivered documentation in an Agile team.

Graduate Teaching Assistant

Mar 2021 – Nov 2021

University of Auckland

- Delivered live code demos for CS719 (Web Technologies: JS, Node.js, HTML, CSS, SQLite) and CS718 (Programming for Industry: Java).

EDUCATION

University of Auckland

Master of Information Technology — First Class Honours (GPA 8.0)

University of Auckland

Postgraduate Certificate in Information Technology — Best Student Award (2020)

South China Agricultural University

Bachelor of Industrial Engineering

CERTIFICATIONS & AWARDS

- AWS Certified Solutions Architect — Associate
- Coursera Deep Learning Specialization (Andrew Ng)
- Google Prompting Essentials
- Prompt Engineering for ChatGPT — Vanderbilt University
- Certificates of Outstanding Achievement: Data Mining · Advanced AI

CORE SKILLS

Agentic AI / LLM	Pydantic-AI · Google Gemini (Flash/Pro) · Claude · Prompt Engineering · Structured Outputs · Tool Use / Web Search · Multi-agent orchestration · RAG · Token optimisation · Async pipelines
Computer Vision	YOLO · SSD · Faster R-CNN · U-Net · Mask R-CNN · SAM · MobileNet · EfficientNet · OpenCV · TensorFlow · PyTorch · Google EdgeTPU / Coral · Vertex AI · AutoML · BigQuery ML
Cloud & Infra	GCP (Cloud Run, Cloud Functions, Cloud Tasks, Pub/Sub, GCS, Secret Manager) · AWS (DynamoDB, AppSync, Lambda, EventBridge) · Terraform · Docker
Backend & Data	Python · asyncio · Pydantic · Flask · FastAPI · REST APIs · Webhooks · SQLAlchemy · Avro · CSV/JSON pipelines · Selenium · BeautifulSoup
Frontend	TypeScript · React 19 · Next.js 16 · Material UI · Real-time WebSocket / AppSync
Practices	Event-driven architecture · Edge quantisation · Concurrency (ThreadPoolExecutor, asyncio) · CI/CD · Agile

REFERENCES

Konstantinos
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